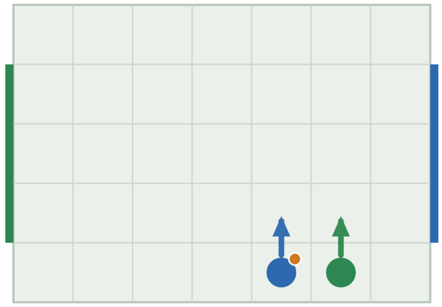
SOCCER MARKOV GAME · RESEARCH MEETING FOLLOW-UP

Player positions and the 4×4 stage game

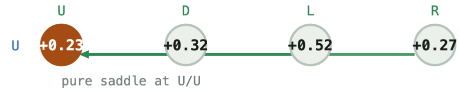
At the research meeting the professor drew a stage game by hand: a small grid of cells, each marked, with arrows chasing each other around the grid to show that no cell is a stable outcome. This is that sketch, generated from the exact solver instead of drawn free-hand, paired with a picture of where the two players actually are, and the exact `4×4 {U, D, L, R}` matrix underneath — the professor's own correction that the matrix, not an entropy number, is the actual output.

A pure state -- one safe cell, nothing to guess

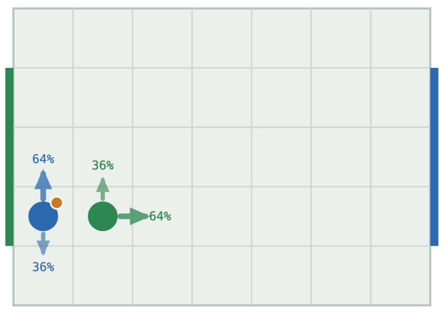


pure equilibrium · V = +0.234

stage game (4, 0, 5, 0, 0)

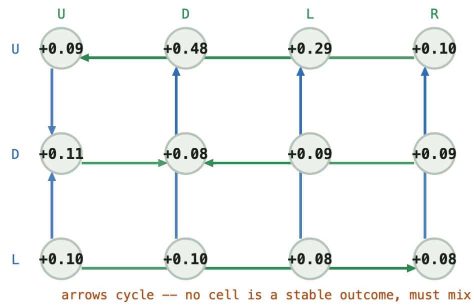


The typical mix -- which goal row to head for

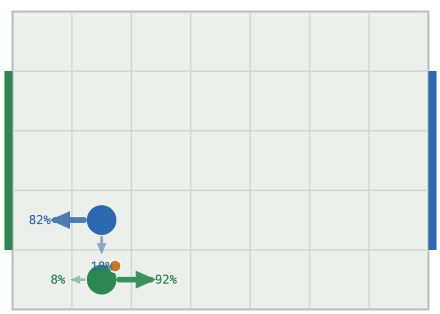


mixed equilibrium · V = +0.094

stage game (0, 1, 1, 1, 0)

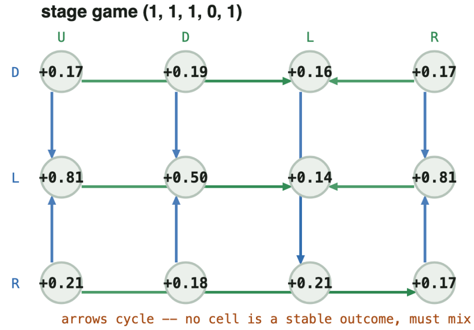


The meeting's example -- indifferent between L and R

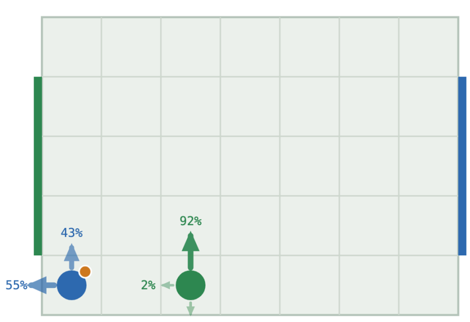


mixed equilibrium · V = −0.206

stage game (1, 1, 1, 0, 1)

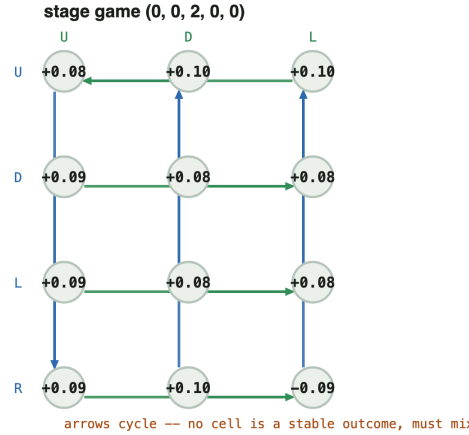


A genuine 3-action mix, not a 2-cycle



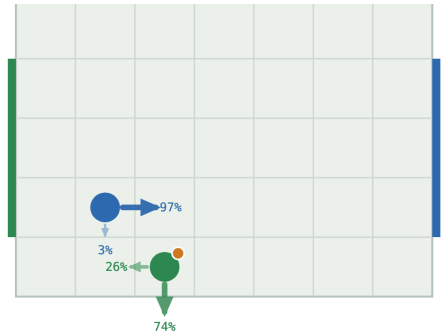
mixed equilibrium · V = +0.085

stage game (0, 0, 2, 0, 0)

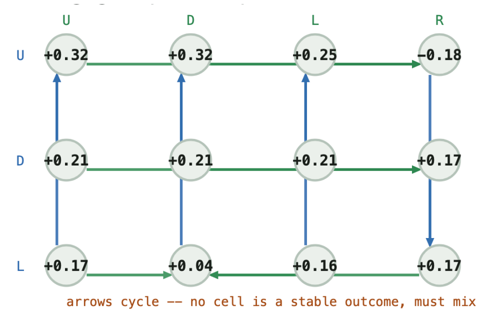


A near-pure hedge -- where rounding would lie

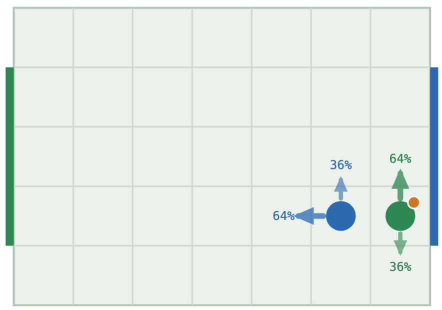
stage game (1, 1, 2, 0, 1)



mixed equilibrium · V = −0.168

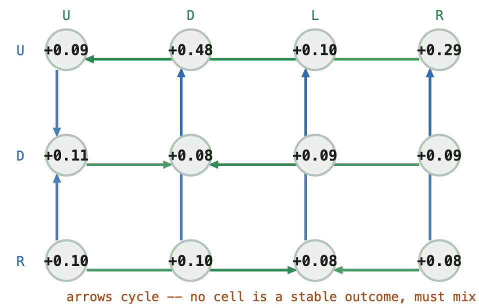


The typical mix, mirrored -- same shape, value negated

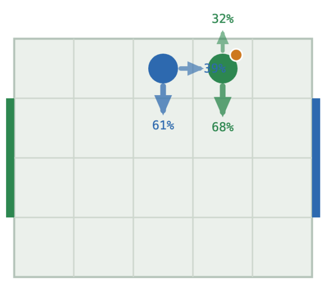


mixed equilibrium · V = −0.094

stage game (5, 1, 6, 1, 1)

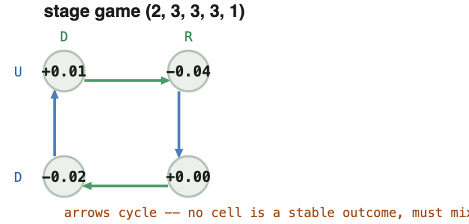


This project's own tackle rule -- a different mechanism, same duel



mixed equilibrium · V = +0.013

stage game (2, 3, 3, 3, 1)



All seven cases: board position (left) and the stage game redrawn as a node-and-arrow graph (right). A lit node with no outgoing arrow is a pure saddle; a closed loop of arrows means no cell is safe.

Reading the graph

Each node is one surviving cell of the stage game after removing strictly dominated rows/columns (`row = the carrier's action`, `col = the defender's action`, always reoriented so the printed payoff is the carrier's, regardless of which player id actually carries the ball). A blue arrow points from a cell to the cell in the same column with the higher payoff —

the carrier's own reason to switch rows. A green arrow points from a cell to the cell in the same row with the *lower* payoff — the defender's reason to switch columns, since the defender minimises.

Case 1 `(4, 0, 5, 0, 0)`

A pure state, for contrast. Support (1,1) — a strict saddle at `U/U`. The carrier plays U always; the defender answers U always; neither has any reason to deviate.

	U	D	L	R
U	0.234	0.317	0.523	0.272
D	0.151	0.211	0.211	0.222
L	0.161	0.182	0.172	0.190
R	0.012	-0.095	0.058	0.098

Case 2 `(0, 1, 1, 1, 0)`

The typical mix. Support (2,2), gap 0.0136. The carrier mixes `U 63.5% / D 36.5%`; the defender answers `U 36.5% / R 63.5%`. This is the common shape: the carrier is deciding which goal row to head for, and the defender is guessing which one. **90 of the 94 mixed states on the canonical board cross a vertical pair exactly like this one.**

	U	D	L	R
U	0.086	0.478	0.286	0.099
D	0.109	0.076	0.085	0.086
L	0.103	0.103	0.085	0.084
R	0.082	0.187	-0.108	-0.071

Case 3 `(1, 1, 1, 0, 1)`

The meeting's own example.

> *"when we move right and when we move left, there's an equal chance of me winning. That's why I am indifferent between the two."*

Support (2,2), gap 0.0445. The carrier's entire live option set is exactly `{L, R}` — no vertical option survives at all, so this is the plain case the meeting asked for: two actions genuinely equally good against the defender's own mix (`D 18% / L 82%`). **This is the rarer shape: only 4 of 94 mixed states cross a purely horizontal pair instead of a vertical one.**

```
       U        D        L        R
U   0.139   -0.173   0.099    0.110
D   0.171    0.185   0.157    0.171
L   0.810    0.500   0.141    0.810
R   0.211    0.181   0.211    0.167
```

Case 4 `(0, 0, 2, 0, 0)`

A genuine 3-action mix. Support (3,3), gap 0.0047, entropy 1.109 bits — the richest mix on this page. Six cells from goal, as far as this board allows, with the defender close enough to threaten three different rows at once: no single row is safely better than the other two. Requested explicitly at the meeting: **the matrix itself, not entropy, is the point.**

```
       U        D        L        R
U   0.084    0.095    0.103    0.110
D   0.086    0.076    0.076    0.086
L   0.086    0.076    0.076    0.086
R   0.088    0.095   -0.088    0.110
```

Case 5 `(1, 1, 2, 0, 1)`

A near-pure hedge, where rounding would lie. Support (2,2) again, but entropy only 0.169 bits — the carrier plays `R 97.5%`, `D 2.5%`. Reading 0.975 as 1.0 would be the rounding mistake the meeting flagged: the gap (0.0043) is real and certified, twelve times smaller than a 0.1 rounding grid would resolve.

	U	D	L	R
U	0.317	0.317	0.247	-0.178
D	0.211	0.211	0.211	0.167
L	0.171	0.045	0.157	0.171
R	0.182	0.182	0.182	0.135

Case 6 `(5, 1, 6, 1, 1)`

The typical mix, mirrored. The board's left-right / role-swap symmetry maps Case 2 exactly onto this state. The values negate to machine precision:

```
V(case 2) = +0.094235
V(mirror) = -0.094235
sum       = -1.39e-17   (zero, to floating-point noise)
```

Direct evidence that Case 2's shape is not a one-off: the identical mix reappears, exactly mirrored, wherever the same relative configuration recurs.

	U	D	L	R
U	0.086	0.478	0.099	0.286
D	0.109	0.076	0.086	0.085
L	0.082	0.187	-0.071	-0.108
R	0.103	0.103	0.084	0.085

Case 7 `(2, 3, 3, 3, 1), 5×4 board`

This project's own tackle rule. Every other case here gets its coin flip from Littman's random move order. This one is solved under the tackle rule (defender commits to a challenge that wins the ball with fixed probability or bounces off) instead — a mechanism with nothing to do with move order at all. **It reduces to the same 2×2 duel shape** (support (2,2), gap 0.0223): the carrier mixes `D 60.7% / R 39.3%`, the defender mixes `U 32.1% / D 67.9%`. Different cause, identical structure.

```
        U        D        L        R
U  -0.012    0.005    0.016   -0.042
D   0.072   -0.022    0.009    0.000
L  -0.019   -0.080    0.019   -0.042
R  -0.007   -0.026   -0.006   -0.039
```

The mathematics behind all seven

A mixed equilibrium is exactly the strategy pair where every action in a player's support earns the **same expected payoff** against the opponent's own mix — that equality is what "indifferent" means in Case 3, and it is also why the best-response graph cycles with no resting point: if one action in the support paid strictly more, that player would raise its weight on it, which is precisely the condition an equilibrium rules out. A best-reply cycle and mutual indifference are the same fact seen from two sides, not two different explanations.

Reproduce with `python scripts/positions.py` (`make positions`) — writes `figures/gallery/positions.svg` and prints every matrix and policy shown here. Full project: `github.com/Charith-Reddy-Pareddy/soccer-markov-nash`.